

4.4 The Mean Value Theorem

The Mean Value Theorem is one of the most important tools in mathematical analysis. We begin this section by stating a theorem that will be crucial in the proof of the Mean Value Theorem.

Theorem (Rolle's Theorem): Let f be a function that satisfies the following three hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$

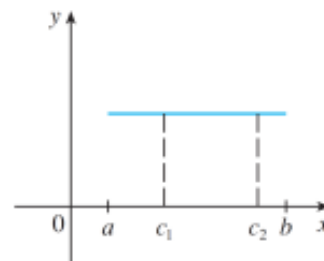
Then there is a number c in (a, b) such that $f'(c) = 0$.

General Idea:

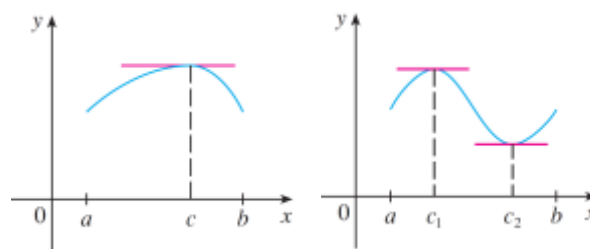
Proof:

We break the proof into three cases.

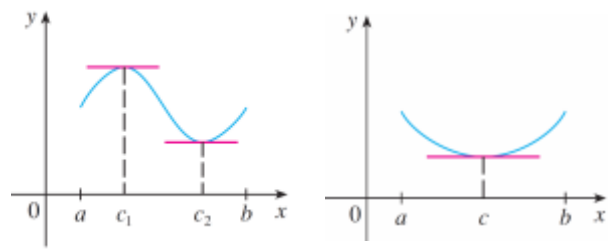
Case 1: $f(x) = k$, a constant.



Case 2: $f(x) > f(a)$ for some x in (a, b) .



Case 3: $f(x) < f(a)$ for some x in (a, b) .



Question: What does Rolle's Theorem tell us about a ball that is thrown upward from ground level and comes back to the ground?

Example 1: Consider the function $f(x) = 5 - 12x + 3x^2$ on $[1, 3]$.

a) Verify that the hypothesis of Rolle's theorem are satisfied on the given interval.

b) Find all values of c in that interval that satisfy the conclusion of the theorem:

Example 2 (if time permits): Use Rolle's Theorem to prove that the equation $x^3 + x - 1 = 0$ has exactly one real solution.

Theorem (The Mean Value Theorem): Let f be a function that satisfies the following hypotheses:

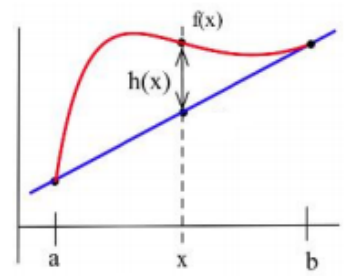
1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .

Then there is a number c in (a, b) such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

General Idea:

Proof:

We begin by constructing an equation for the secant line AB .



Example 3: Verify that the function $f(x) = x^3 - x$ satisfies the conditions of the Mean Value Theorem on the interval $[0, 2]$. Then find all numbers c that satisfy the conclusion of the Mean Value Theorem.

Question: If $f'(x) = 0$ for all x in (a, b) , what can we say about the function f ?

Theorem: If $f'(x) = 0$ for all x in (a, b) , then f is constant on (a, b) .

Corollary: If $f'(x) = g'(x)$ for all x in (a, b) , then there is a constant C such that $f(x) = g(x) + C$ for every x in (a, b) .

Example 4: Use the most recent theorem to prove the most recent corollary.
(Hint: Consider the function $F(x) = f(x) - g(x)$)